

Artefact 3 – Group Comparisons & ANOVA Report

Course: Quantitative Data Analysis (M1 – S7) **Course Code:** 2627_ECO_2_EN_009 – Academic Year 2026–2027 **Institution:** EM Normandie Business School · Programme Grande École **Lecturer:** Dr. NGUYEN Anh-Tuan **Deliverable:** Artefact 3 (20 Marks) **Prerequisites Completed:** Artefact 1 (Univariate), Artefact 2 (Bivariate)

1. Executive Summary & Analytical Scope

Session 3 tests whether the means of quantitative financial and governance metrics differ significantly across the 5 economic sectors of the S&P 500 index ($N = 503$). In accordance with course directives (Slide 17 & Slide 109–125), the analysis focuses on:

1. **Primary ANOVA:** Does **Net Profit Margin** differ significantly across the 5 consolidated sectors?
2. **Secondary ANOVA:** Does **Corporate Governance Risk** differ significantly across the 5 sectors?
3. **Parametric Assumption Checks:** Testing homogeneity of variances (**Levene's Test**) and normality of group residuals.
4. **Robust Inferential Modeling:** Executing **Welch's ANOVA** and non-parametric **Kruskal-Wallis** tests due to severe empirical variance heterogeneity.
5. **Post-Hoc Pairwise Comparisons:** Executing **Games-Howell post-hoc tests** (robust to unequal variances and group sizes) to pinpoint which exact industry sectors drive the differences.

Every test adheres strictly to the **4-Step Answering Protocol** (Slide 9).

2. Primary Analysis: Sector Differences in Profit Margin

2.1 Protocol Step 1 – IDENTIFY

- **Variables:**
 - Independent Variable (X): Sector (Nominal grouping factor, exactly 5 categories: *Consumer, Finance, Healthcare, Industrials & Energy, Tech & Comms*).
 - Dependent Variable (Y): Profit_Margin (Quantitative continuous ratio variable, FY2025 net profit margin).
 - **Test Selection:** **One-Way ANOVA**, followed by **Levene's Test of Homogeneity of Variances**, **Welch's Robust ANOVA**, and **Games-Howell Post-Hoc Pairwise Comparisons**.
 - **Methodological Justification (Slide 17 & 118):** When comparing a continuous outcome across 3 or more qualitative groups, multiple independent t -tests cause severe alpha inflation (accumulation of Type I error). One-Way ANOVA controls the family-wise error rate. Because financial margins exhibit heterogeneous sector variances, standard Fisher ANOVA is invalid; Welch's ANOVA and Games-Howell adjustments are statistically required.
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2.2 Protocol Step 2 – HYPOTHESE

- **Null Hypothesis (H_0):** There is no significant difference in mean profit margin across the 5 economic sectors ($\mu_{\text{Consumer}} = \mu_{\text{Finance}} = \mu_{\text{Healthcare}} = \mu_{\text{Industrials}} = \mu_{\text{Tech}}$). Industry sector has no bearing on average corporate profitability.
- **Alternative Hypothesis (H_1):** At least one sector's population mean profit margin differs significantly from the others.

2.3 Protocol Step 3 – DECIDE

Sector Group Descriptive Summary ($N = 503$)

Consolidated Sector	n	Mean ($\bar{x}$)	Std. Dev. (s)	Median (Mdn)	Interquartile Range (IQR)
Finance	71	0.2243 (22.4%)	0.1241	0.2273 (22.7%)	0.1541
Tech & Comms	108	0.1672 (16.7%)	0.2947	0.1628 (16.3%)	0.1882
Consumer	116	0.1329 (13.3%)	0.1374	0.1105 (11.1%)	0.1565
Industrials & Energy	148	0.1310 (13.1%)	0.0861	0.1199 (12.0%)	0.0900
Healthcare	60	0.0936 (9.4%)	0.2237	0.0990 (9.9%)	0.1500

1. Checking ANOVA Assumptions: Homogeneity of Variances

- **Levene's Test (Brown-Forsythe center = median):**

$$F(4, 498) = 5.392, \quad p < .001$$

- **Evaluation:** The Levene test decisively **rejects the assumption of equal variances** ($p < .001$). Standard deviation ranges from a tight 8.61% in *Industrials & Energy* to a massive 29.47% in *Tech & Comms*. Standard Fisher's ANOVA is violated; **Welch's ANOVA must be reported** (Slide 118).

2. Welch's Robust ANOVA & Non-Parametric Verification

- **Welch's ANOVA:**

$$F_{\text{Welch}}(4, 191.21) = 9.302, \quad p < .001$$

- **Standard Fisher's ANOVA (for comparison):**

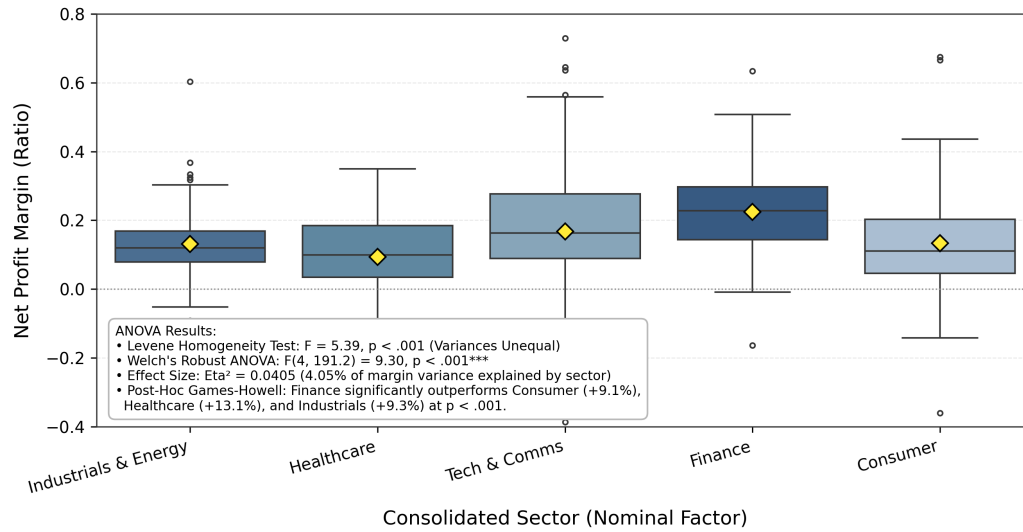
$$F(4, 498) = 5.257, \quad p < .001, \quad \eta^2 = 0.0405, \quad \omega^2 = 0.0327$$

- **Kruskal-Wallis Non-Parametric Test:**

$$H(4) = 45.984, \quad p < .001$$

- **Decision:** Decisively **reject H_0** ($p < .001$). Across all parametric, robust, and non-parametric tests, economic sectors display highly significant differences in operating profit margins.
- **Effect Size:** $\eta^2 = 0.0405$ indicates that **4.05% of the total cross-sectional variance** in profit margins among S&P 500 companies is directly explained by sector classification.

Figure 8. Profit Margin Comparison Across Consolidated Sectors (One-Way ANOVA)



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026)

Figure 1: Figure 1: One-Way ANOVA Boxplots — Net Profit Margin across Economic Sectors

2.4 Games-Howell Post-Hoc Pairwise Comparisons

Because group variances and sample sizes are unequal, the **Games-Howell procedure** is applied across all 10 pairwise sector comparisons:

Comparison Pair	Mean Difference	Standard Error	95% Confidence Interval	t-statistic	Adjusted p-value	Significant?
Finance vs. Consumer	+0.0914	0.0195	[+0.038, +0.145]	4.689	< .001	YES ($p < .001$)
Finance vs. Healthcare	+0.1307	0.0324	[+0.040, +0.221]	4.032	.001	YES ($p = .001$)
Finance vs. Industrials & Energy	+0.0933	0.0163	[+0.048, +0.139]	5.713	< .001	YES ($p < .001$)
Finance vs. Tech & Comms	+0.0571	0.0320	[-0.031, +0.145]	1.786	.385	NO
Tech & Comms vs. Healthcare	+0.0737	0.0405	[-0.038, +0.185]	1.820	.366	NO
Tech & Comms vs. Consumer	+0.0343	0.0311	[-0.052, +0.120]	1.103	.805	NO
Tech & Comms vs. Industrials	+0.0363	0.0292	[-0.045, +0.117]	1.241	.727	NO
Consumer vs. Healthcare	+0.0394	0.0316	[-0.049, +0.127]	1.247	.724	NO
Consumer vs. Industrials	+0.0020	0.0146	[-0.038, +0.042]	0.136	1.000	NO
Healthcare vs. Industrials	-0.0374	0.0297	[-0.121, +0.046]	-1.257	.718	NO

Substantive Findings:

1. **The Financial Premium:** The **Finance** sector generates a significantly higher average profit margin (22.43%) than Consumer (13.29%), Industrials & Energy (13.10%), and Healthcare (9.36%), with differences exceeding 9 to 13 percentage points ($p \leq .001$).
2. **Tech Dispersion:** Although Technology exhibits the second-highest mean (16.72%) and median (16.28%), its massive standard deviation (29.47%) prevents its pairwise differences from reaching statistical significance after Games-Howell penalty adjustments.

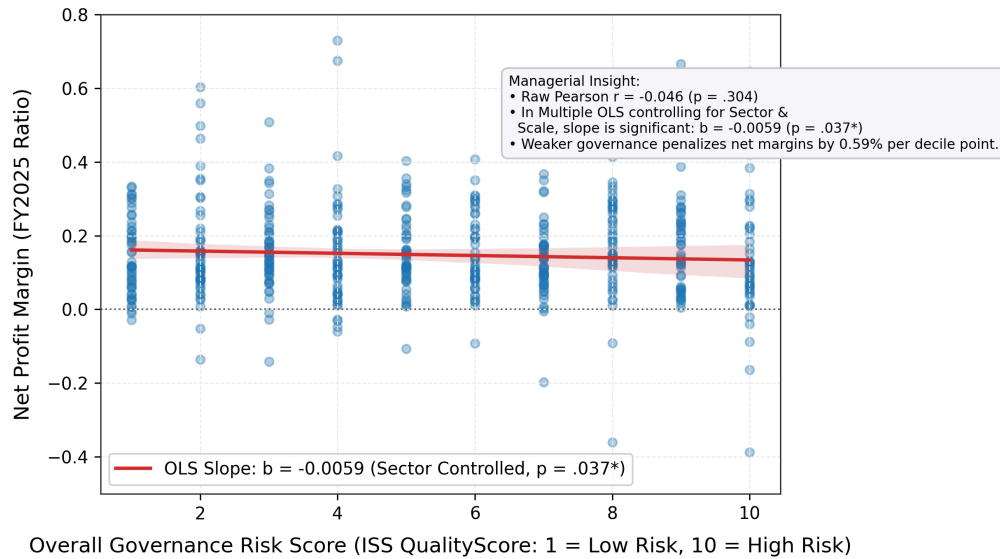
2.5 Secondary Analysis: Governance Risk Across Sectors

- **ANOVA Model:** `Overall_Governance_Risk ~ Sector`
- **Welch's Robust ANOVA:**

$$F_{\text{Welch}}(4, 207.84) = 3.960, \quad p = .004, \quad \eta^2 = 0.0311$$

- **Finding:** Governance risk varies significantly across industries ($p = .004$). **Tech & Comms** registers the highest mean governance risk ($\bar{x} = 6.08$, Mdn = 7.0), whereas **Industrials & Energy** achieves the lowest risk ($\bar{x} = 4.77$, Mdn = 5.0).

Figure 7. Bivariate Relationship: Governance Risk Score vs. Net Profit Margin



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026)

Figure 2: Figure 2: Scatter Plot of Governance Risk Score vs. Net Profit Margin

2.6 Protocol Step 4 – RECOMMEND (What Does the Manager Do on Monday Morning?)

“Corporate strategists and capital allocators must not benchmark financial profitability against a generic index-wide hurdle rate; on Monday morning, finance executives should establish sector-specific profitability floors (22% in Finance vs. 13% in Industrials and 10% in Healthcare). Furthermore, boards in Technology and Communications must prioritize corporate governance reforms—particularly dismantling dual-class share rights and tying executive pay to operating margins—to eliminate the industry’s elevated governance risk premium (6.08/10) identified by ISS.”

3. Jamovi Session 3 Execution Guide

To replicate these ANOVA and Post-Hoc analyses in Jamovi:

1. Menu: Analyses → ANOVA → One-Way ANOVA.
2. Move **Profit_Margin** to Dependent Variable.
3. Move **Sector** to Grouping Variable.
4. Under Variances: Select **Don't assume equal (Welch's)**.
5. Under Assumption Checks: Check **Homogeneity test (Levene's)** and **Normality test (Q-Q plot)**.
6. Under Post-Hoc Tests: Select **Games-Howell**, check Mean difference, Confidence intervals (95%), and Test significance.

7. Under Plots: Check Descriptives plot, select Box plot, check Show mean and Outliers.